



Analog IC Design

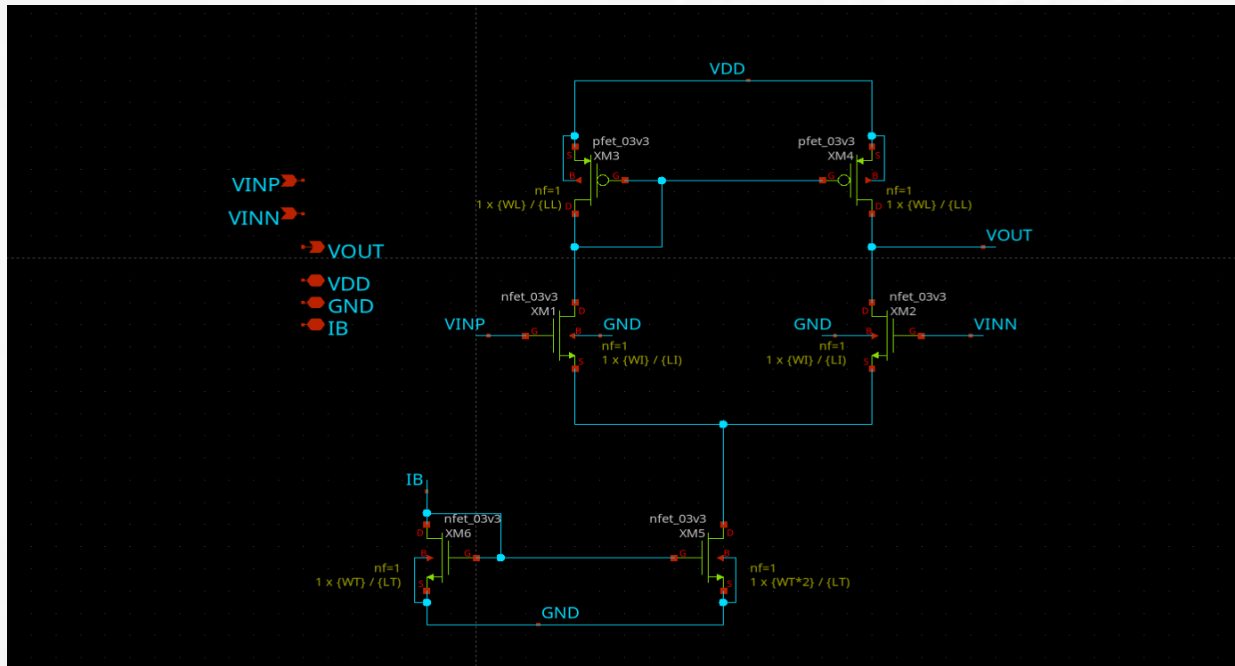
CMOS Technology

MOHAMMED NASR EL-DIN

Section 3 - Lab 07

5T OTA design

Sizing



Parameter	Value
Supply voltage	1.8V
Load	5pF
Open loop DC voltage gain	$\geq 34\text{dB}$
CMRR @ DC	$\geq 74\text{dB}$
Phase margin (PM)	$\geq 70^\circ$
Current seed (reference current)	10uA
CM input range – low	$\leq 1.2\text{V}$
CM input range – high	$\geq 1.5\text{V}$
GBW	$\geq 5\text{MHz}$
OTA current consumption	Minimize
Area	Minimize

From the CMIR spec you need to identify the input pair type: NMOS or PMOS?

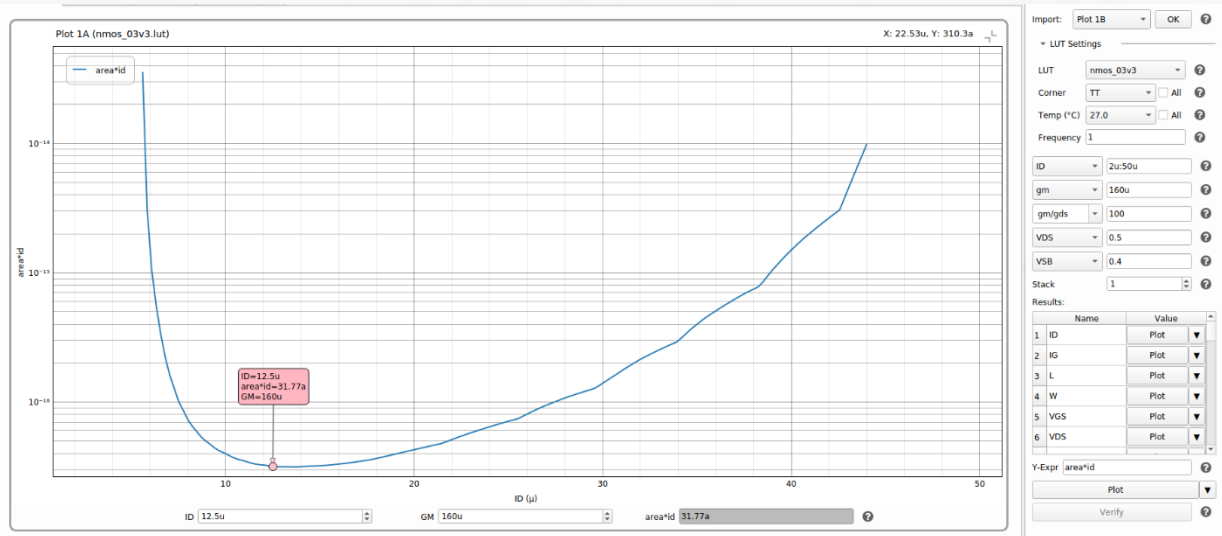
Since the CMIR is closer to V_{DD} rail, we need to use **NMOS** input pair.

Let $V_{DS3} = 0.9v$ "diode connected", $V_{DS1} = 0.5v \Rightarrow V_{DS5} = 1.8 - V_{DS3} - V_{DS1} = 0.4v$, $V_{SB1} = V_{DS5} = 0.4v$

Current selection

We have to select the current that can provide minimum Area and power consumption and meet all the specs.

So, we can get the minimum value of the current at the minimum area by plotting the next graph.



Log-scale Area $\times I_D$ vs I_D

The area is **minimized** as the current **increase** till certain value then start increasing with the current.

The current range that can provide minimum area: **$10\mu A \rightarrow 17\mu A$**

Taking good value that can be getting from tail current source by simple mirroring ratio m.

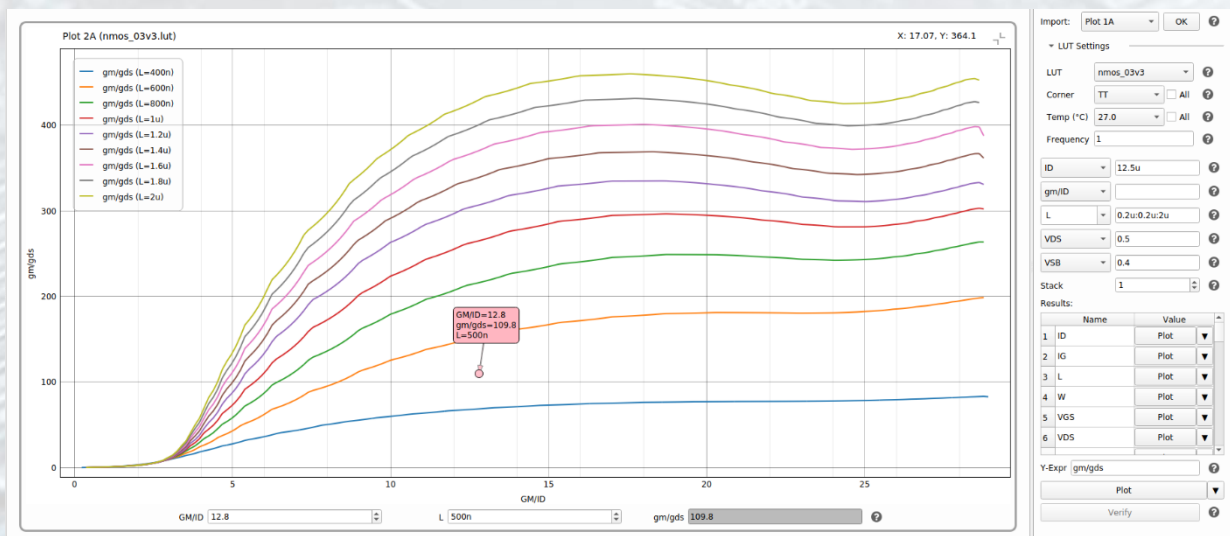
let $m = \frac{5}{2} = 2.5 \Rightarrow I_D = 10\mu A \times \frac{2.5}{2} = 12.5\mu A$.

NMOS input pair design

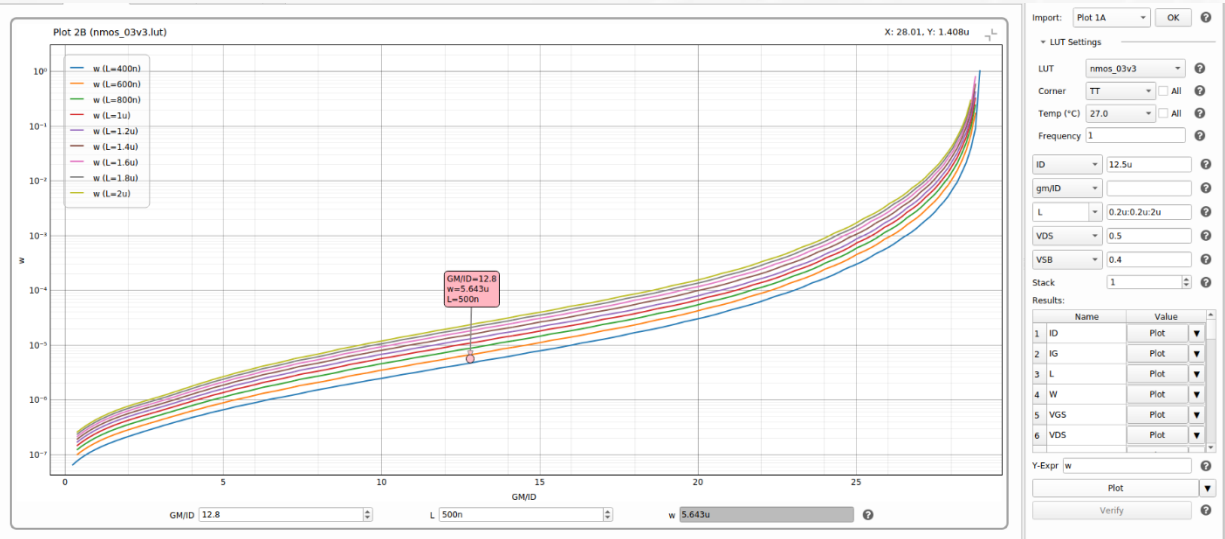
$$V_{DS1} = 0.5v, V_{SB1} = V_{DS5} - 0 = 0.4v, I_D = 12.5\mu A$$

$$GBW = \frac{g_{m1}}{2\pi C_L} \geq 5MHz, C_L = 5pF \Rightarrow g_{m1} \approx 160\mu S \Rightarrow \frac{g_{m1}}{I_D} = \frac{160\mu S}{12.5\mu A} = 12.8\mu S/\mu A$$

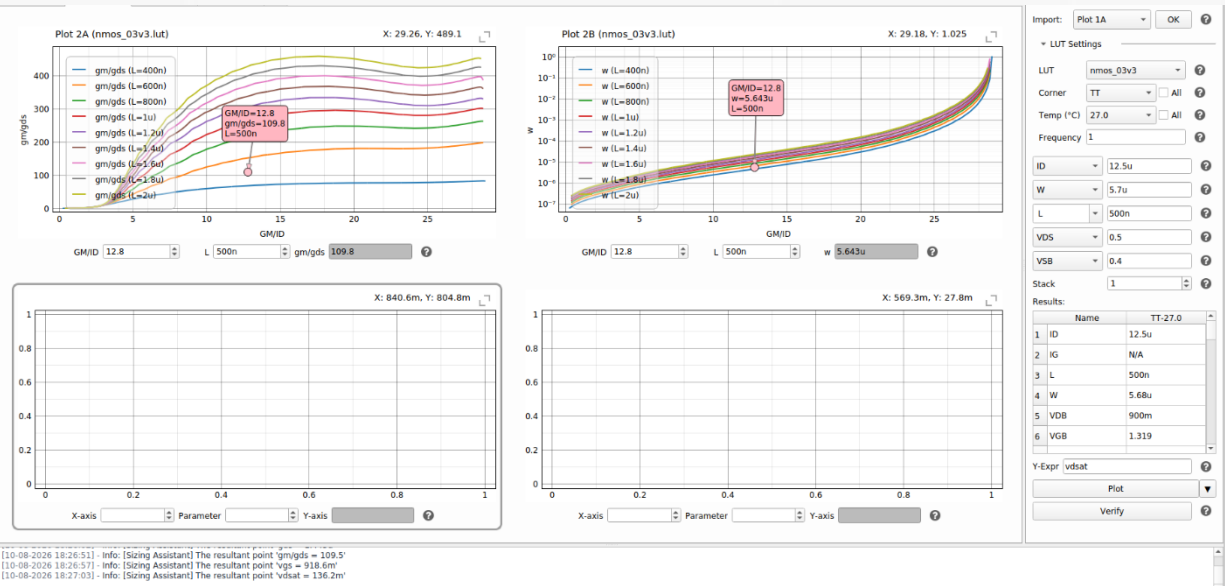
$$\text{Let } g_{ds1} = g_{ds3} \Rightarrow \text{gain} = \frac{g_{m1}}{2g_{ds1}} = 34dB \approx 50 \Rightarrow \frac{g_{m1}}{g_{ds1}} = 100 \Rightarrow g_{ds1} = 1.6\mu S$$



$$\frac{g_m}{g_{ds}} \text{ vs } \frac{g_m}{I_D}: L = 500nm \text{ at } \frac{g_m}{g_{ds}} = 109.8$$



$W \text{ vs } \frac{g_m}{I_D}: W \approx 5.7\mu\text{m} \text{ at the selected } L$



[10-08-2026 18:26:53] Info: [Sizing Assistant] The resultant point 'gm/ids' = 109.5'

[10-08-2026 18:26:57] Info: [Sizing Assistant] The resultant point 'vgs' = 918.6m'

[10-08-2026 18:27:03] Info: [Sizing Assistant] The resultant point 'vdsat' = 136.2m'

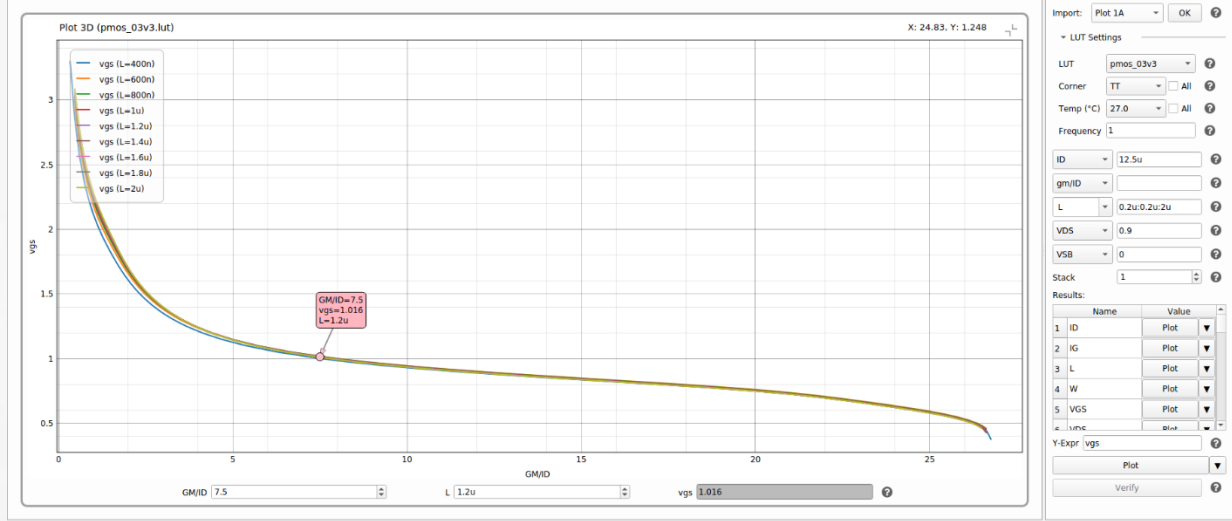
Input design point

PMOS active load design

$$V_{DS3} = 0.9v, V_{SB3} = 0, I_D = 12.5\mu A$$

$$V_{DD} = V_{gs3} + V_{dsat1} - V_{gs1} + V_{icm} \Rightarrow V_{icm} = V_{DD} - V_{dsat1} + V_{gs1} - V_{gs3} \geq 1.5v$$

$$\Rightarrow 1.8 - 0.1 + 0.9 - V_{gs3} \geq 1.5 \Rightarrow V_{gs3} \leq 1.1v, V_{gs3max} = 1.1v$$

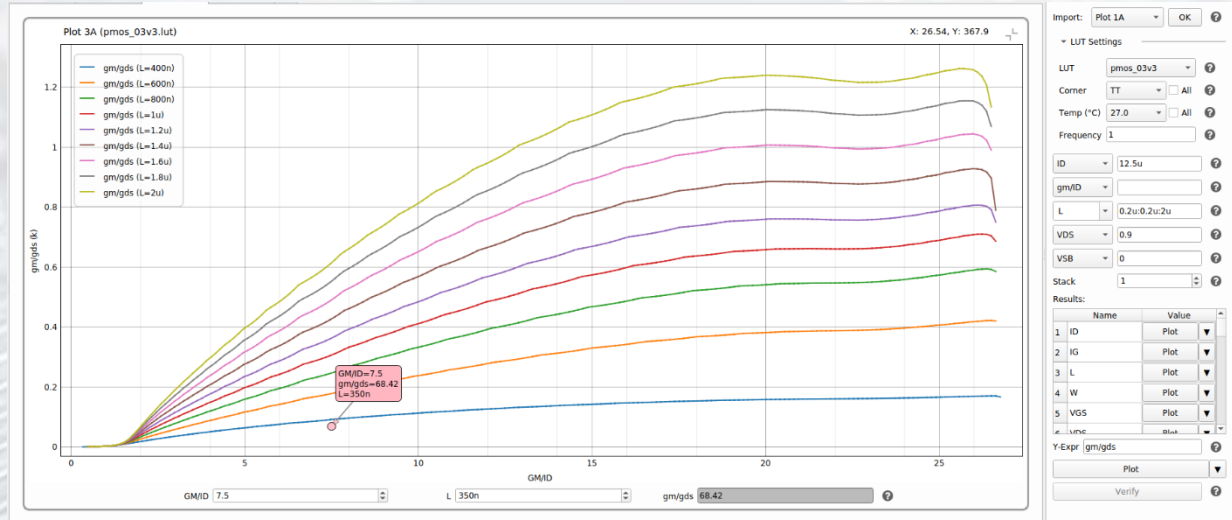


$$V_{gs} \text{ vs } \frac{g_m}{I_D} : \text{independent on the length}$$

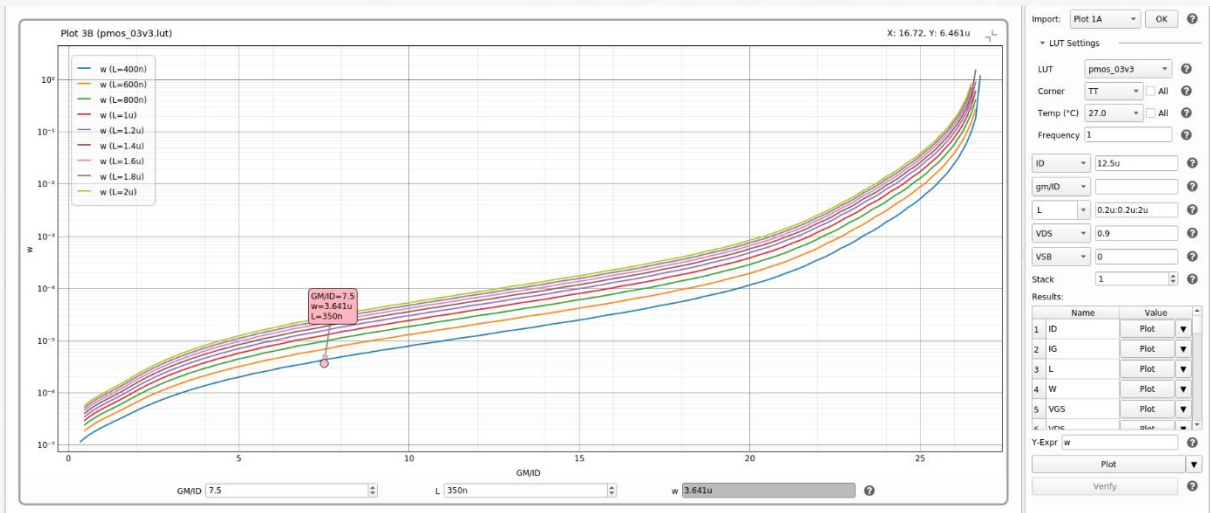
Biasing the transistor in strong inversion region (small $\frac{g_m}{I_D}$) to get small g_m and consequentially **minimized** Area.

$$\text{Take } \frac{g_{m3}}{I_D} = 7.5\mu S/\mu A \text{ at } V_{gs3} = 1.016 < 1.1v$$

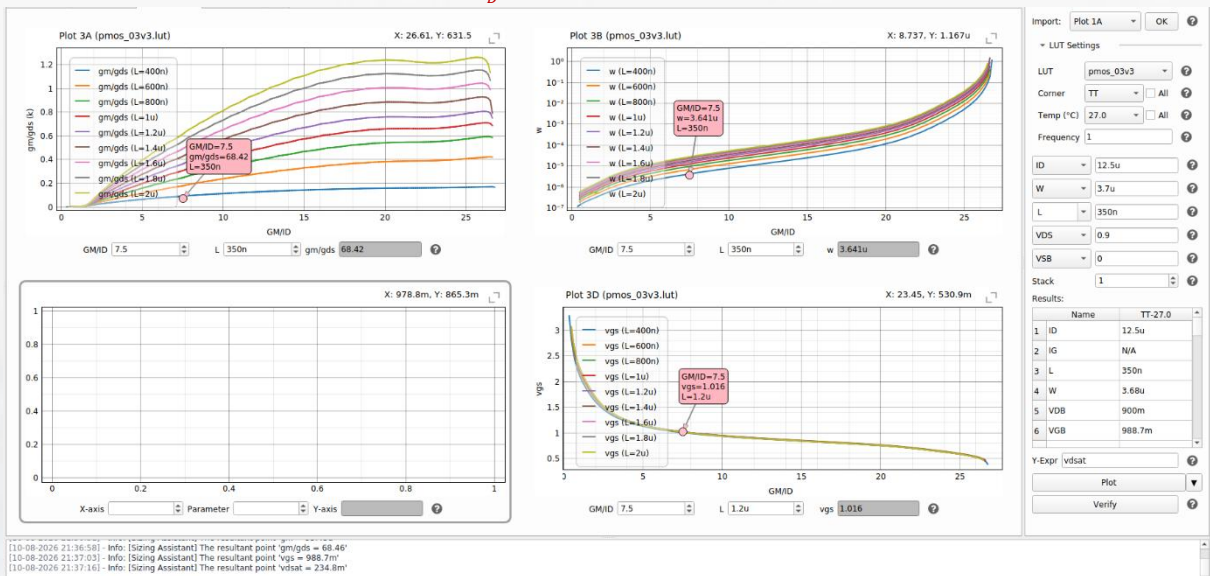
$$\Rightarrow g_{m3} = 93.75\mu S, \text{ let } g_{ds3} = g_{ds1} = 1.6\mu S \Rightarrow \frac{g_{m3}}{g_{ds3}} = \frac{93.75}{1.6} = 58.6$$



$$\frac{g_m}{g_{ds}} \text{ vs } \frac{g_m}{I_D} : L = 350nm \text{ at } \frac{g_m}{g_{ds}} = 68.4$$



W vs $\frac{g_m}{I_D}$: $W \approx 3.7\mu m$ at the selected L

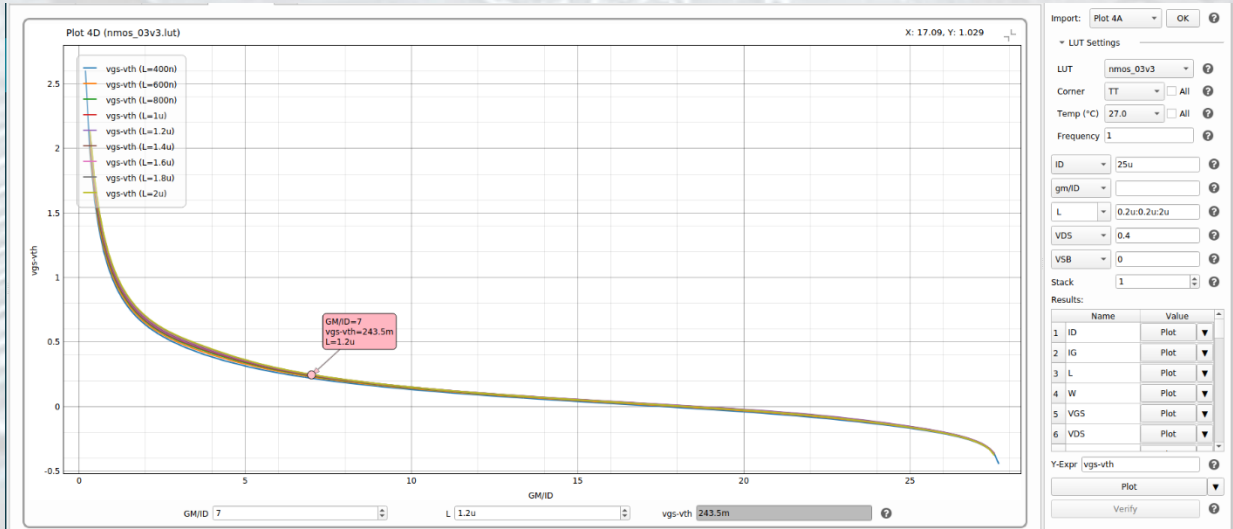


Load design point

NMOS Tail CS design

$$V_{DS5} = 0.4v, V_{SB3} = 0, I_D = 25\mu A$$

$$V_{icm} = V_{gs1} - V_{th5} + V_{gs5} \leq 1.2v \Rightarrow V_{gs5} - V_{th5} \leq 1.2 - V_{gs1} \approx 0.3v \Rightarrow V_{gs5} \leq (V_{gs5} - V_{th5})_{max} = 0.3v$$



$V_{gs} - V_{th}$ vs g_m/I_D : independent on the length

Biassing the transistor in strong inversion region (small $\frac{g_m}{I_D}$) to get small g_m and consequentially **minimized** Area.

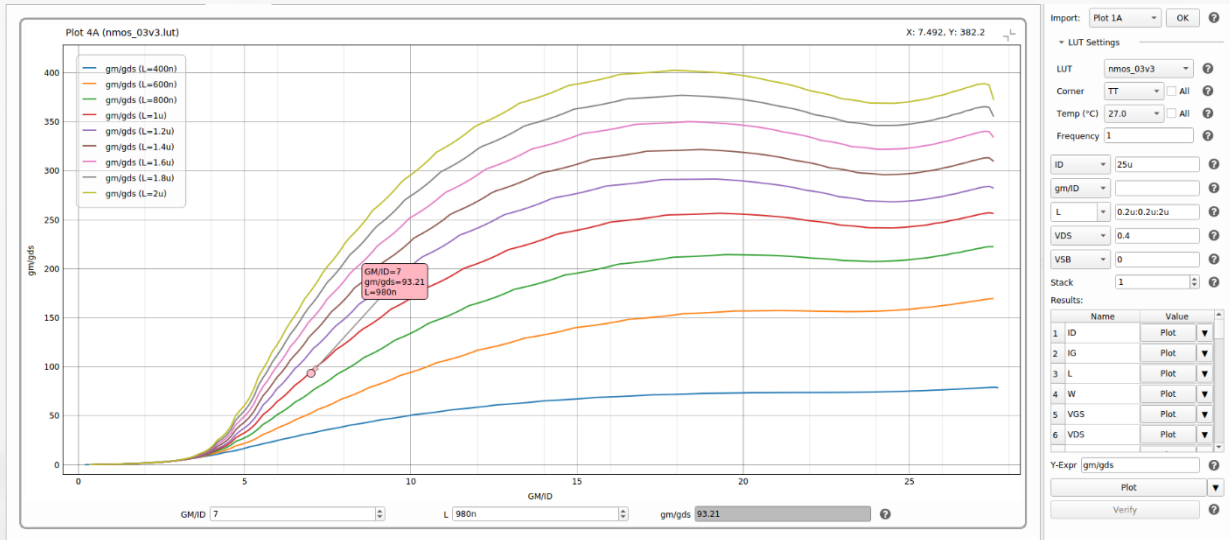
Take $\frac{g_{m5}}{I_D} = 7$ at $V_{gs5} - V_{th5} \approx 0.245V < 0.3V$

$$\Rightarrow g_{m5} = 175\mu S$$

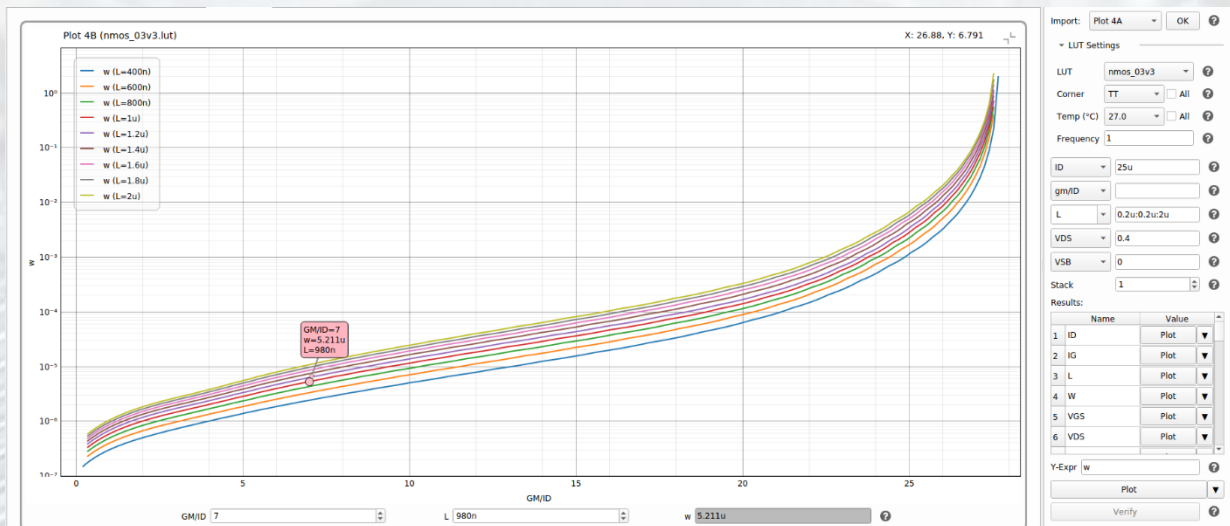
$$CMRR = \frac{A_{vd}}{A_{vcm}} \geq 74dB \Rightarrow A_{vcm} = A_{vd} - CMRR = 34dB - 74dB = -40dB$$

$$A_{vcm} = \frac{1}{2g_{m3}R_{ss}} = 10^{-\frac{40}{20}} \Rightarrow g_{ds5} = \frac{1}{R_{ss}} = 2g_{m3} \times 10^{-\frac{40}{20}} = 2 \times 93.75\mu S \times 10^{-\frac{40}{20}} \approx 1.9\mu S$$

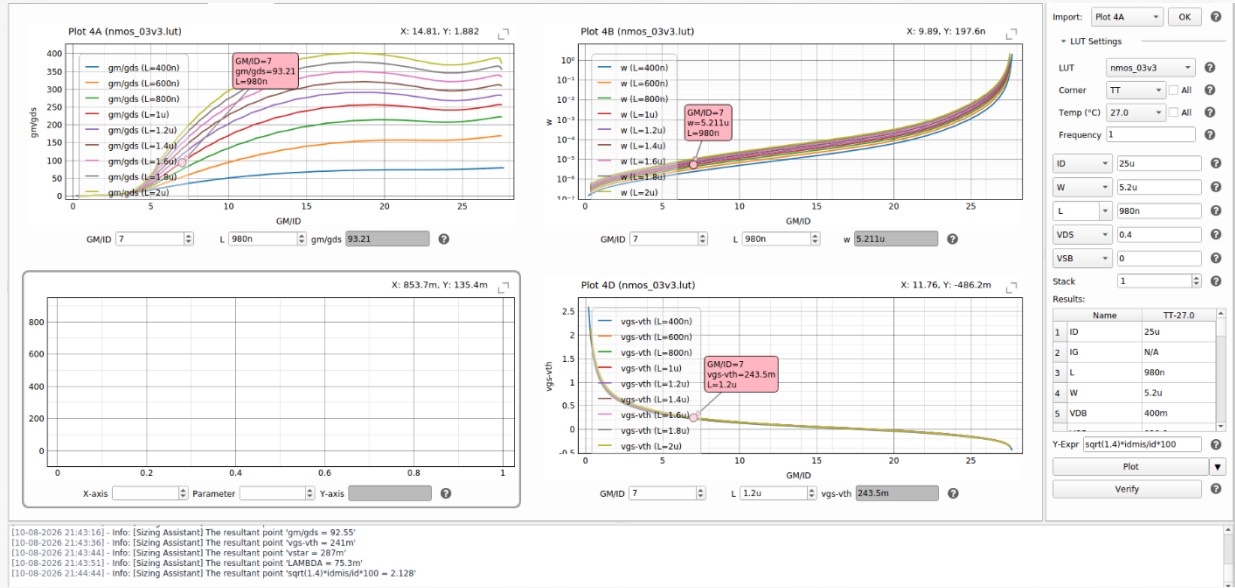
$$\Rightarrow \frac{g_{m5}}{g_{ds5}} = 92.1$$



$$\frac{g_m}{g_{ds}} \text{ vs } \frac{g_m}{I_D} : L = 980nm \text{ at } \frac{g_m}{g_{ds}} = 93.2$$



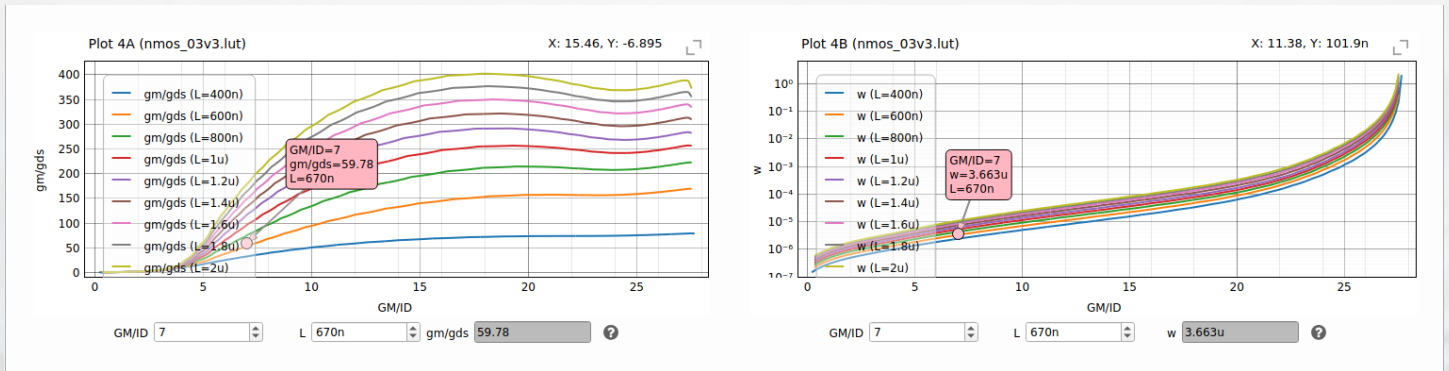
$$W \text{ vs } \frac{g_m}{I_D} : W \approx 5.2\mu m \text{ at the selected } L$$



Tail CS design point

The above result show that at the design point the CM has $\lambda = 0.075 < 0.1$ and mismatch $\sigma \approx 2.1\%$ which are good results.

Note: after simulation the selected sizing I found that it gives excess gain and CMRR, so I reduced the tail current source to minimize the area.



Sizing tuning: $L = 670nm, W = 3.75\mu m$

$$W_6 = \frac{W_5}{2.5} \approx \frac{3.75\mu m}{2.5} = 1.5\mu m$$

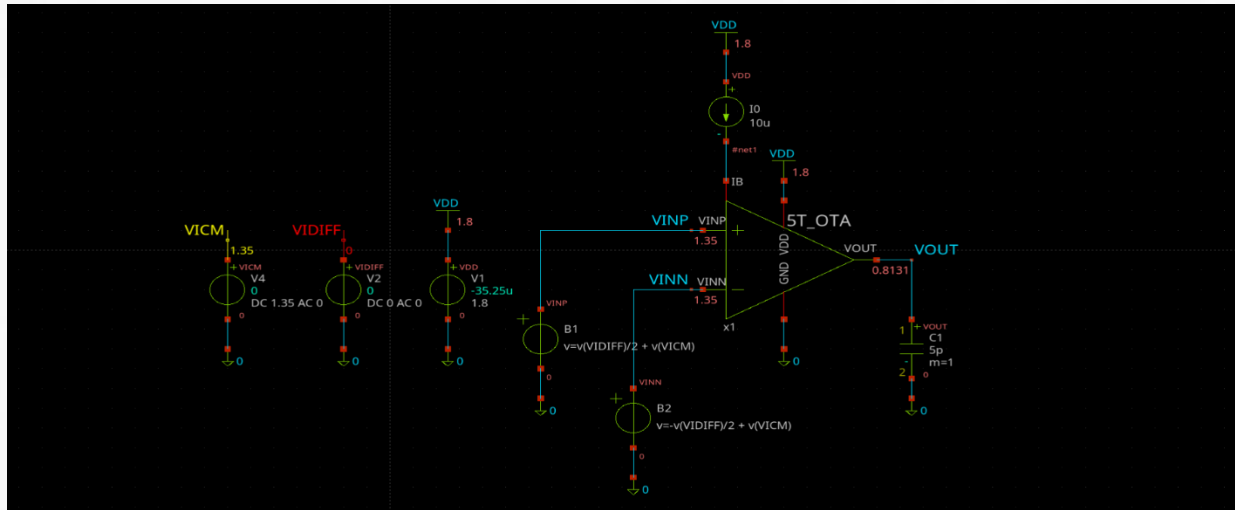
M_X	$W(\mu m)$	$L(nm)$	$g_m(\mu S)$	$I_D(\mu A)$	$\frac{g_m}{I_D} (\frac{\mu S}{\mu A})$	$V^*(mv)$	$V_{dsat}(mv)$	$V_{ov}(mv)$
$M_{1,2}$	5.7	500	158.2	12.5	12.65	158	136.2	81.25
$M_{3,4}$	3.7	350	93.45	12.5	7.5	268	235	225
$M_{5,6}$	3.75 – 1.5	670	176.5	25 – 10	7	283	236.5	230

The above results are getting from ADT

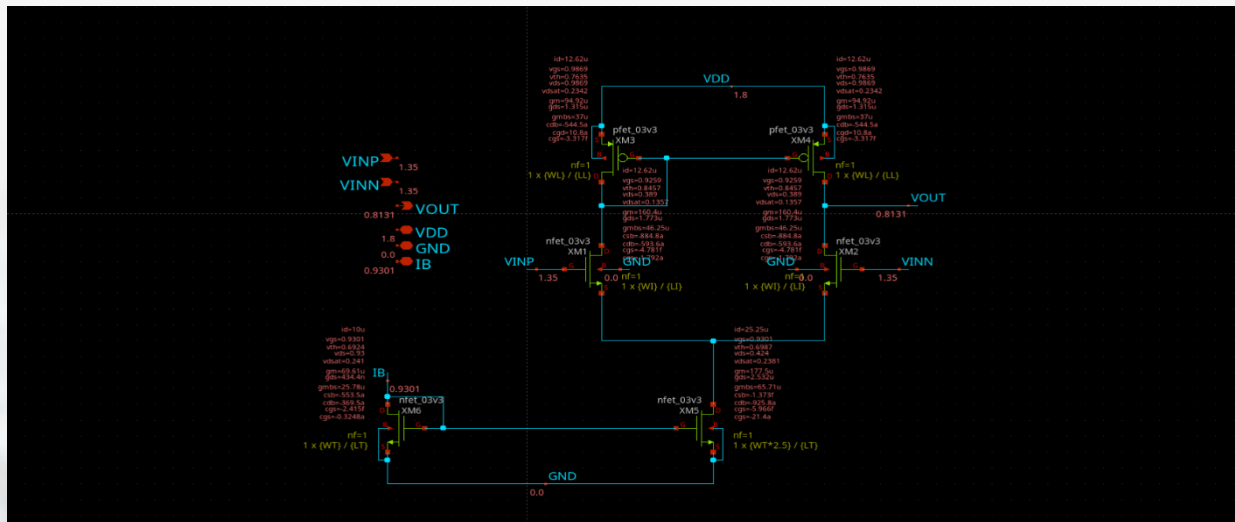
$$\text{Total area: } 2 \times 5.7\mu m \times 0.5\mu m + 2 \times 3.7\mu m \times 0.35\mu m + (1 + 2.5) \times 1.5\mu m \times 0.67\mu m = 11.8\mu m^2$$

Open-Loop OTA Simulation

Op analysis



Total current = 35.25 μ A



```

=== MOSFET Operating Points ===
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm4.m0      m.x1.xm3.m0      m.x1.xm6.m0
model       pfet_03v3.8      pfet_03v3.8      pfet_03v3.9
id          1.26229e-05  1.26229e-05  1e-05
vgs         0.986892    0.986892    0.930057
vds         0.986888    0.986888    0.930049
vth         0.763455    0.763455    0.692368
vdsat       0.234245    0.234245    0.241026
gm          9.4918e-05   9.4918e-05   6.96085e-05
gds        1.31524e-06   1.31524e-06   4.34436e-07
gmbs       3.69982e-05     3.69982e-05   2.57839e-05

BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm5.m0      m.x1.xm2.m0      m.x1.xm1.m0
model       nfet_03v3.9      nfet_03v3.9      nfet_03v3.9
id          2.52459e-05   1.26229e-05  1.26229e-05
vgs         0.930057    0.925942    0.925942
vds         0.424039    0.389042    0.389042
vth         0.698668    0.845723    0.845723
vdsat       0.238091    0.135703    0.135703
gm          0.00017752  0.000160408  0.000160408
gds        2.53192e-06  1.77256e-06  1.77256e-06
gmbs       6.57149e-05     4.62468e-05   4.62468e-05

=== Bias Points ===
M1: gm/id = 12.7076 S/A, Vstar = 0.157386 V
M2: gm/id = 12.7076 S/A, Vstar = 0.157386 V
M3: gm/id = 7.51949 S/A, Vstar = 0.265976 V
M4: gm/id = 7.51949 S/A, Vstar = 0.265976 V
M5: gm/id = 7.03166 S/A, Vstar = 0.284428 V
M6: gm/id = 6.96085 S/A, Vstar = 0.287321 V
    
```

Is the current (and gm) in the input pair exactly equal? **Yes**, they are

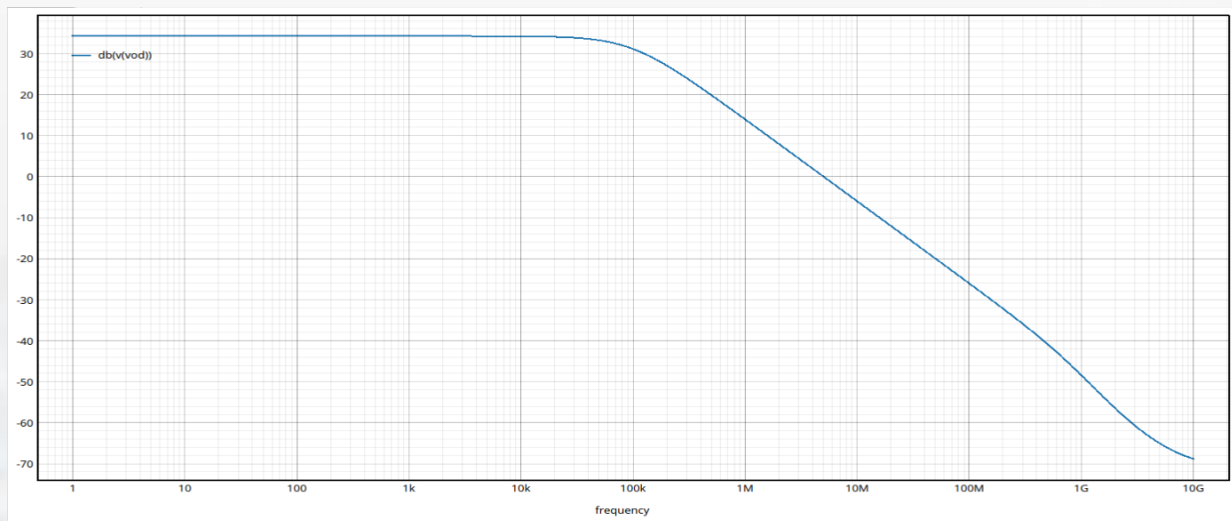
What is DC voltage at VOUT? Why? $V_{O-DC} = 0.81v$, almost as $V_F = 1.8 - V_{gs3}$ due to **internal feedback**

AC analysis

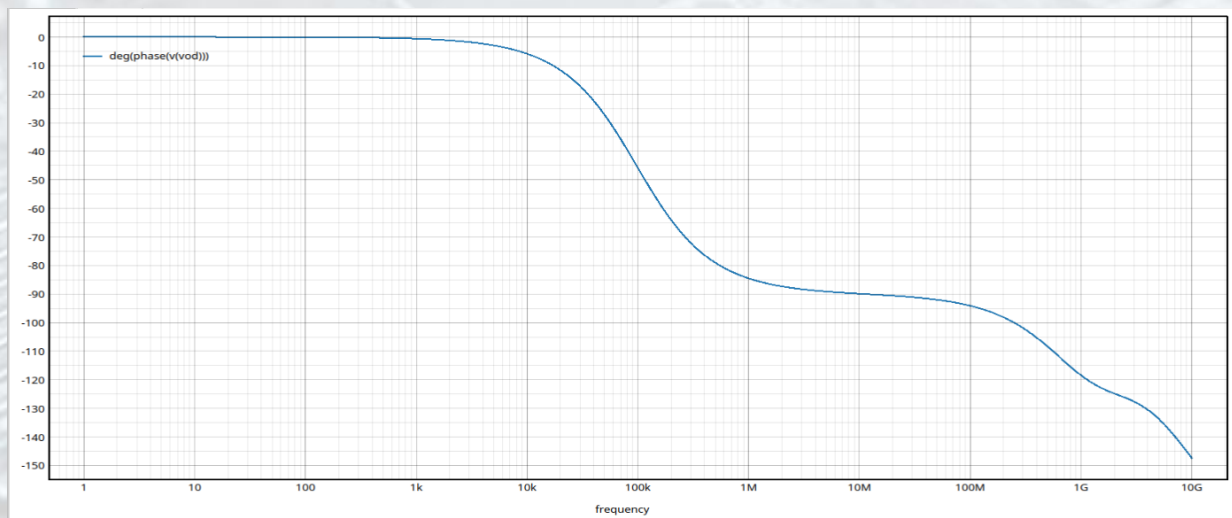
```
av_diff_db      = 3.425149e+01
av_diff_mag     = 5.159107e+01
bw_diff        = 9.721698e+04
ugf            = 5.014530e+06
p_ugf          = -8.909808e+01
av_cm_db       = -3.994657e+01
av_cm_mag      = 1.006170e-02
bw_cm          = 9.721914e+04

=== AC Results ===
Av Diff: 34.2515 dB, BW: 97217 Hz
UGF: 5.01453E+06 Hz
GBW: 5.01553E+06 Hz
PM: 90.9019 deg
Av CM: -39.9466 dB, BW: 97219.1 Hz
CMRR: 74.1981 dB
```

Diff small signal



Bode plot: A_{vd} magnitude(dB)

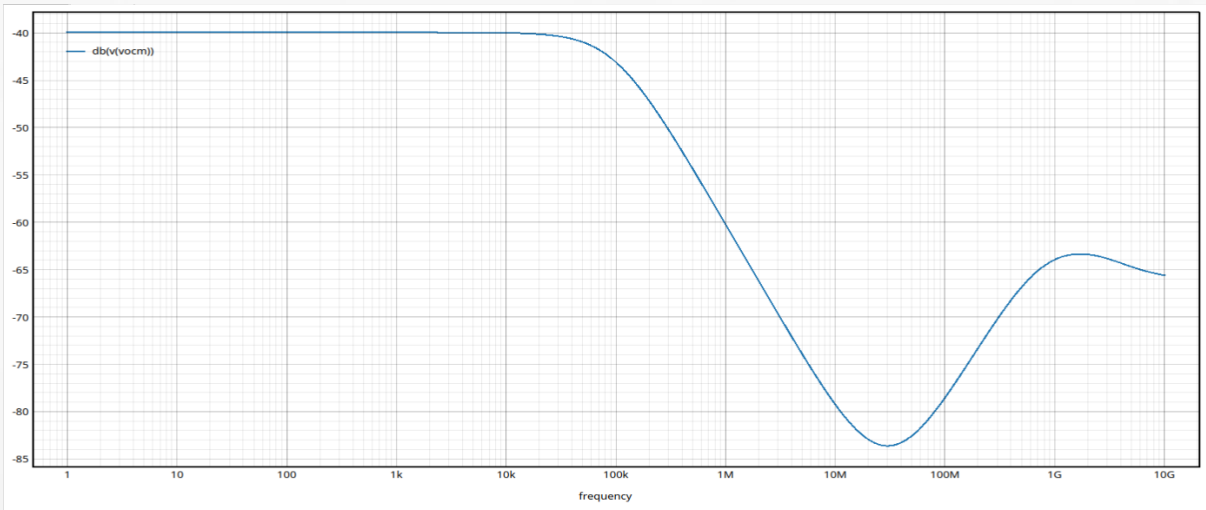


Bode plot: A_{vd} phase

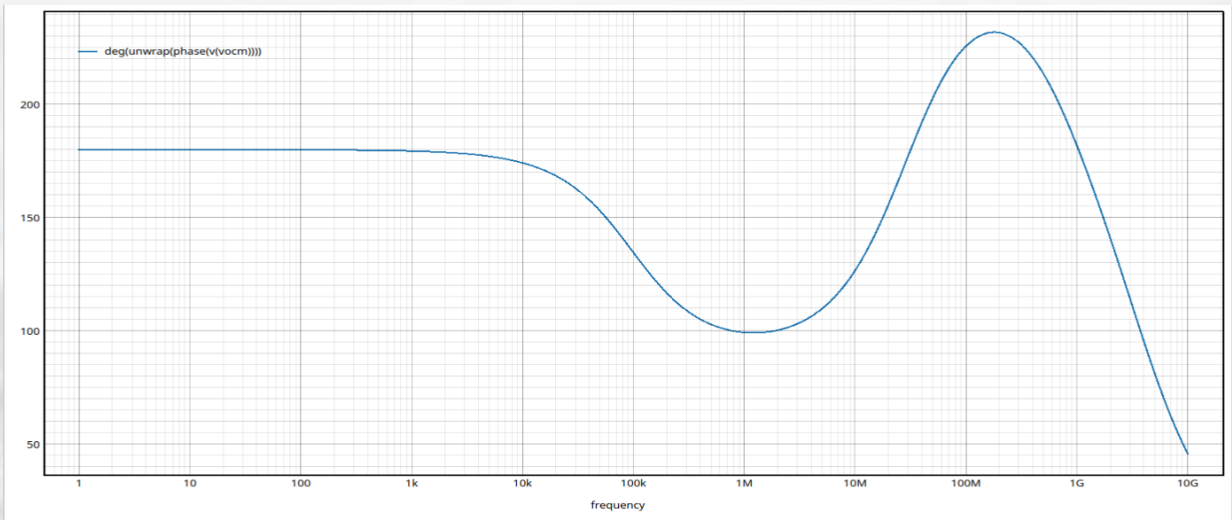
Compare simulation results with hand calculations in a table.

parameter	Simulation	Hand calculation
$ A_{vd} $	$51.6 \equiv 34.25dB$	$50 \equiv 34dB$

CM small signal



Bode plot: A_{vcm} magnitude(dB)

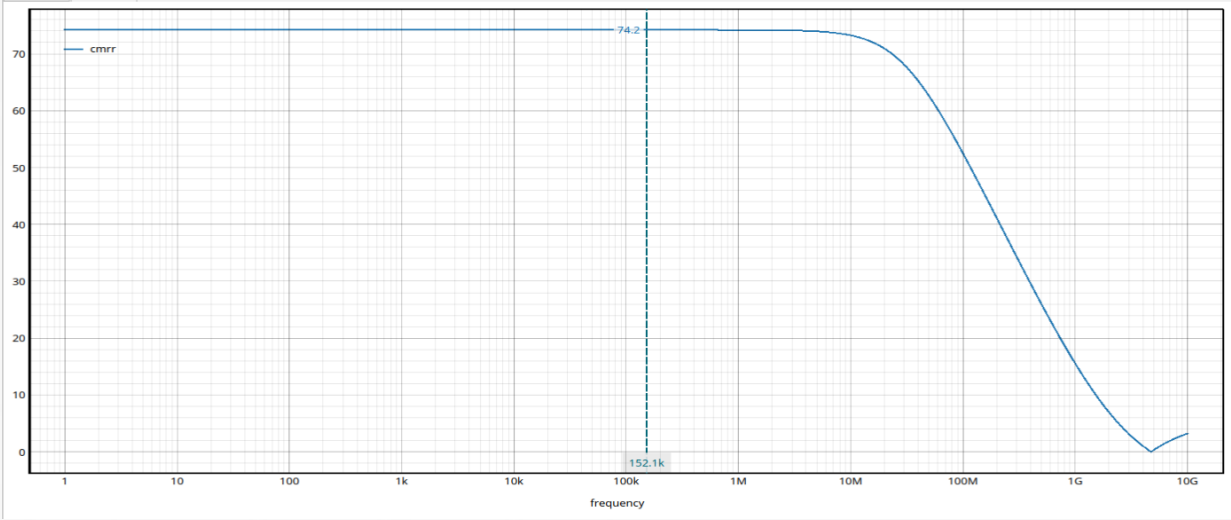


Bode plot: A_{vcm} phase

Compare simulation results with hand calculations in a table.

parameter	Simulation	Hand calculation
$ A_{vcm} $	$0.01 \equiv -40dB$	$0.01 \equiv -40dB$

CMRR

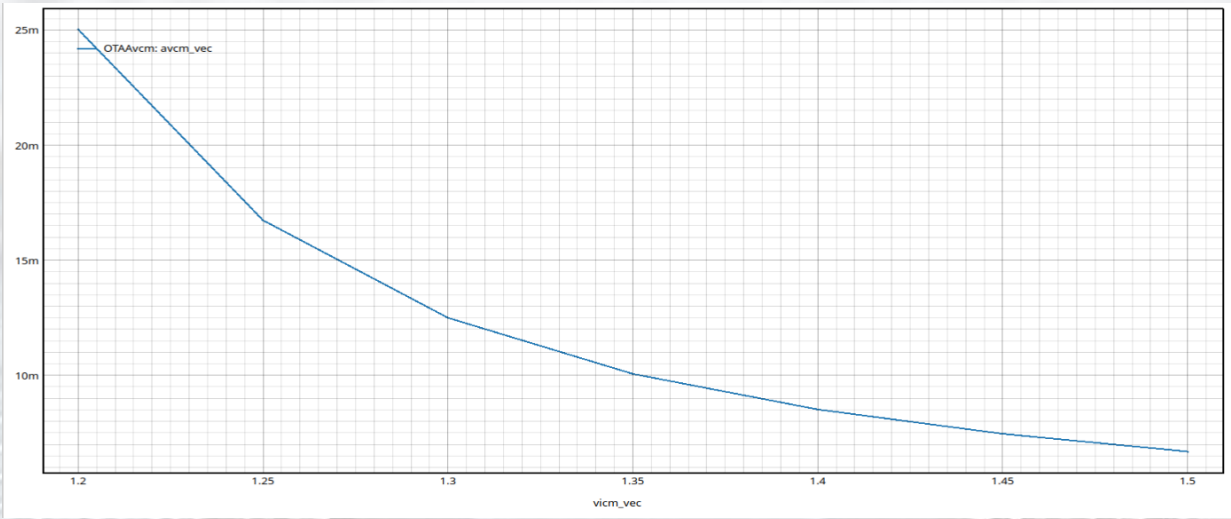


CMRR

Compare simulation results with hand calculations in a table.

parameter	Simulation	Hand calculation
CMRR	74.2dB	74dB

A_{VCM} vs V_{ICM}

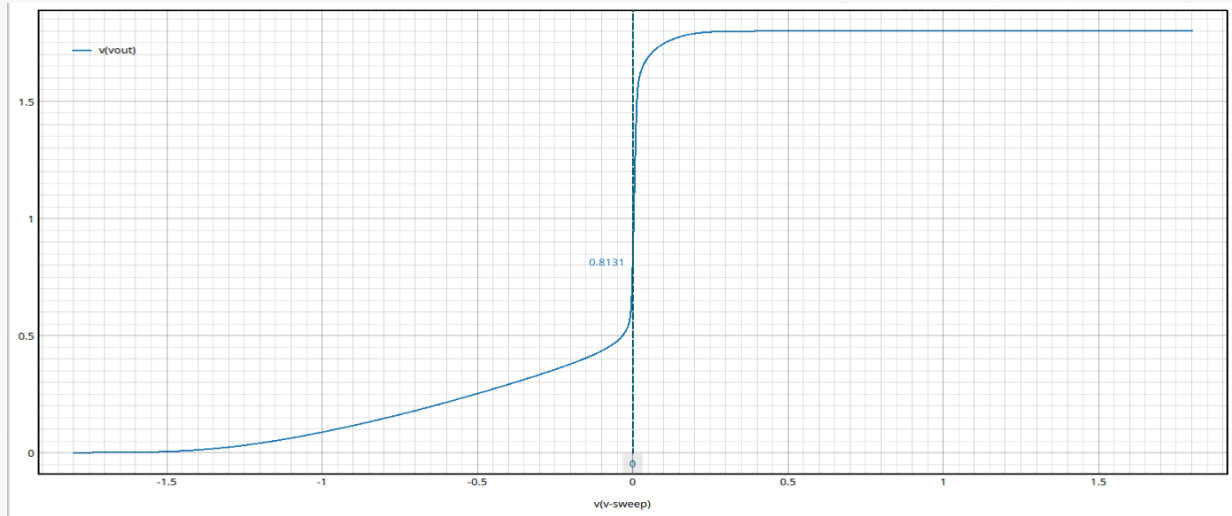


Justify the results. Does A_{VCM} increase or decrease vs V_{ICM} ? Why?

A_{vcm} decrease with V_{icm} , since as V_{icm} increase the voltage on the tail current source will increase and consequentially R_{ss} will increase, $\Rightarrow \left(A_{vcm} \propto \frac{1}{R_{ss}}\right)$ decrease.

DC analysis

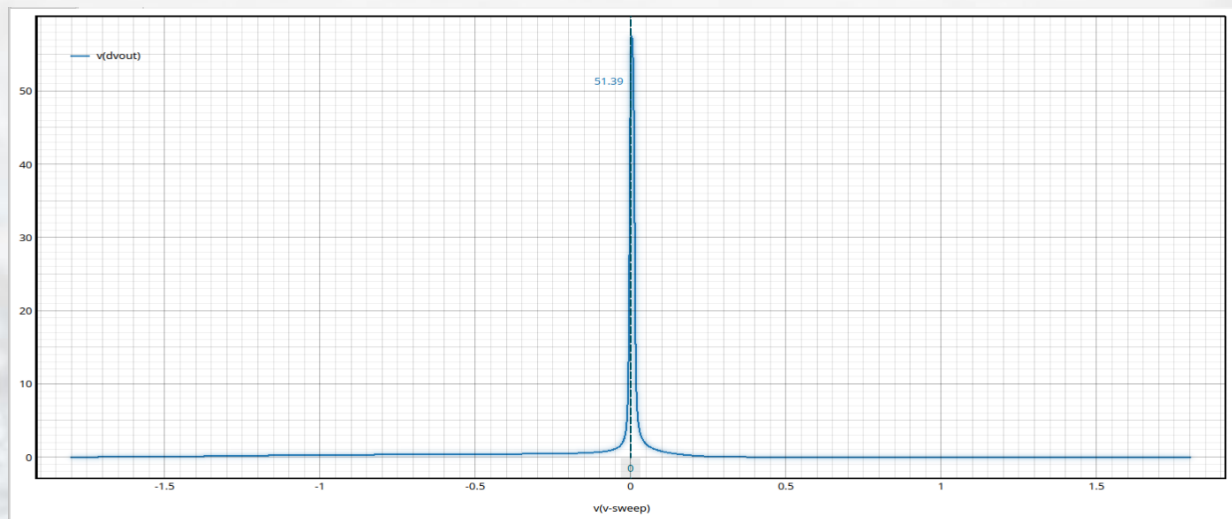
Diff large signal



V_{out} vs V_{id}

From the plot, what is the value of V_{out} at $V_{id} = 0$? Why?

The value of V_{out} at $V_{id} = 0$ is the common mode value since $V_{out} = V_{ocm} + V_{od} = 0.813 + V_{id}A_{vd} = 0.813$ at $V_{id} = 0$



Derivative of V_{out} vs V_{id}

Compare the peak with A_{vd} . Comment.

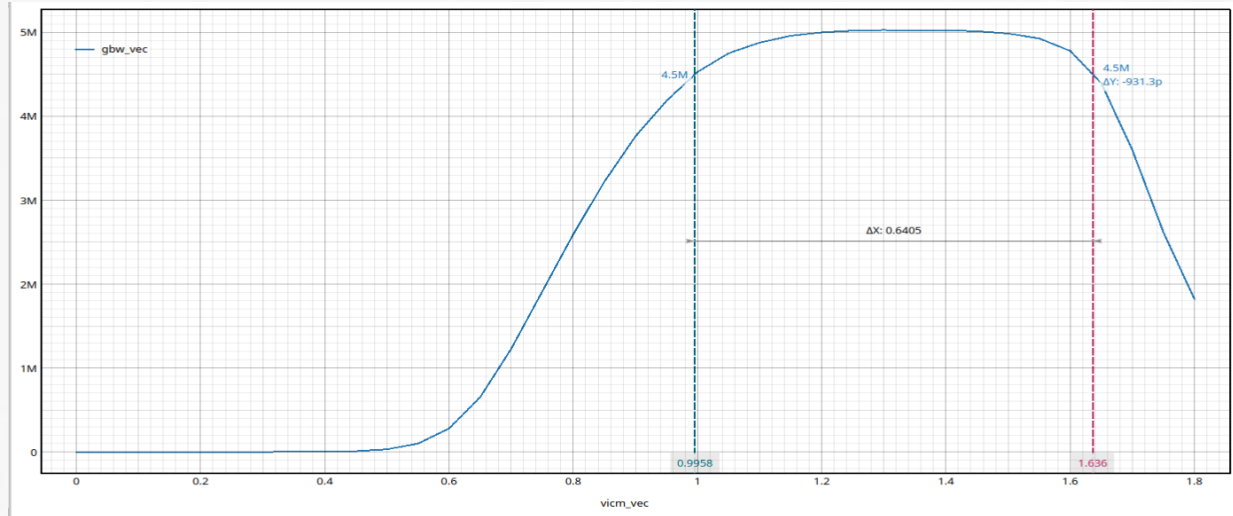
The peak same as the small signal gain $A_{vd} = \frac{dV_{od}}{dV_{id}}$

As V_{id} increase, V_{INN} decrease and V_{out} increase and saturate at VDD.

As V_{id} decrease, V_{INN} increase and V_{out} decrease $\Rightarrow M_2$ goes out of saturation first then M_5 , $\Rightarrow V_{out}$ decrease slowly then saturate close to GND.

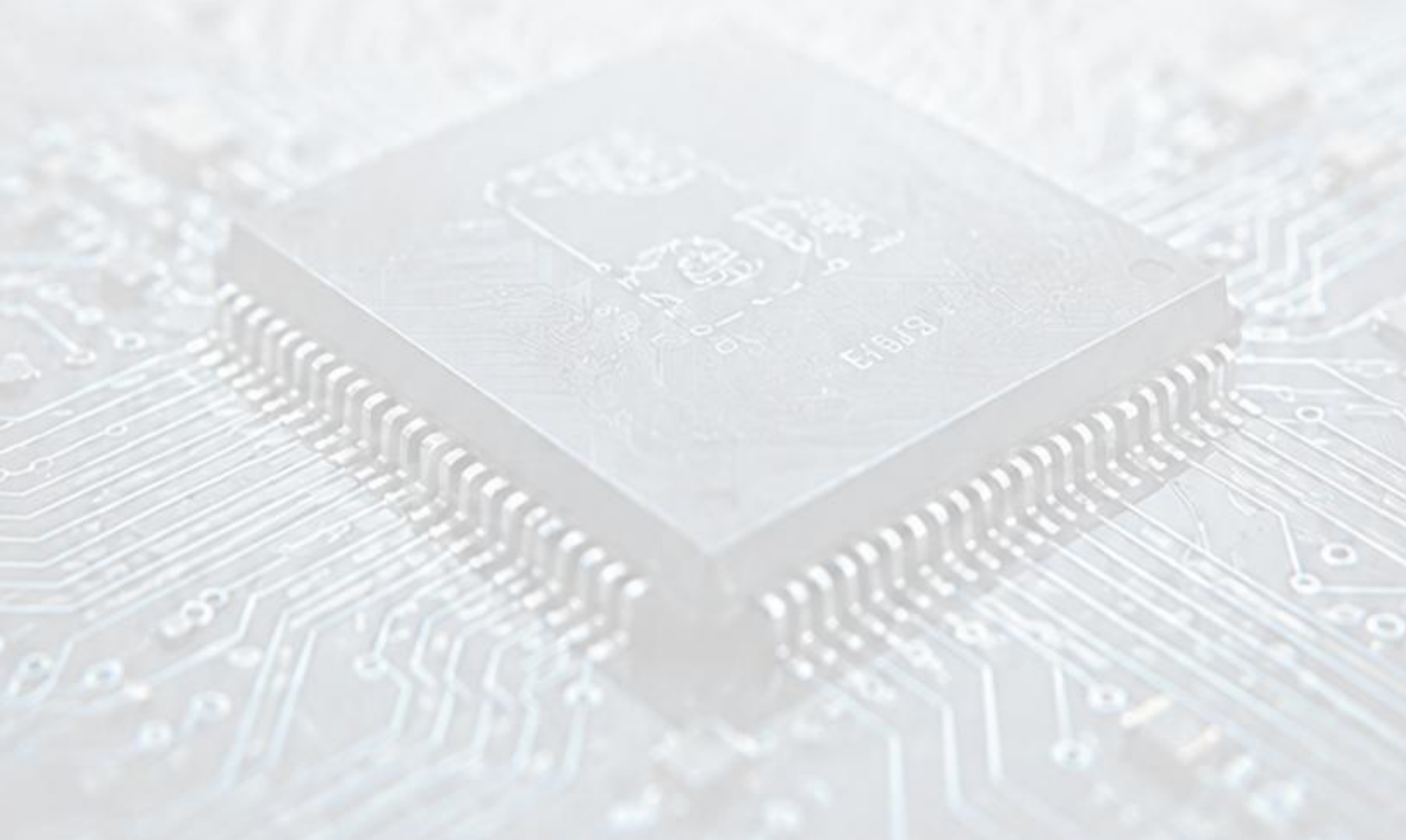
Diff Large Signal Analysis Results:
Max Output: 1.8 V
Min Output: 0.000123307 V

CM large signal (GBW vs VICM)



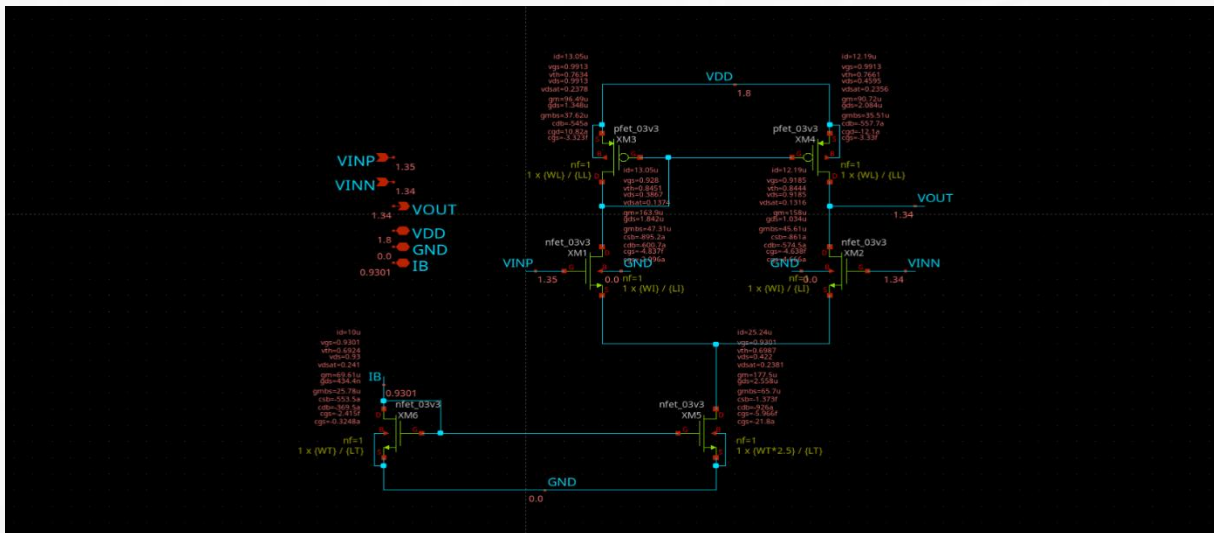
GBW vs V_{icm} with 10% acceptable degradation

$$\min V_{icm} \approx 1v, \max V_{icm} \approx 1.64v \Rightarrow CMIR(1v \rightarrow 1.64v) = 0.64v$$



Closed-Loop OTA Simulation

OP analysis



```

=== MOSFET Operating Points ===
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm4.m0      m.x1.xm3.m0      m.x1.xm6.m0
model       pfet_03v3.8      pfet_03v3.8      nfet_03v3.9
id          1.2193e-05  1.30477e-05      1e-05
vgs         0.99127    0.99127          0.930057
vds         0.459523   0.991265         0.930049
vth         0.766142   0.763433         0.692368
vdsat       0.235614   0.237796         0.241026
gm          9.07197e-05  9.64866e-05      6.96085e-05
gds         2.08428e-06   1.3485e-06       4.34436e-07
gmbs        3.55119e-05   3.76201e-05      2.57839e-05

BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm5.m0      m.x1.xm2.m0      m.x1.xm1.m0
model       nfet_03v3.9      nfet_03v3.9      nfet_03v3.9
id          2.52406e-05   1.21933e-05      1.30477e-05
vgs         0.930057   0.918464         0.927995
vds         0.421986   0.918461         0.386718
vth         0.69868    0.8444           0.845144
vdsat       0.238091    0.131637         0.137447
gm          0.000177467   0.00015804       0.000163933
gds         2.55813e-06   1.03401e-06      1.84175e-06
gmbs        6.56956e-05   4.56114e-05      4.73053e-05

=== Bias Points ===
M1: gm/id = 12.5641 S/A, Vstar = 0.159183 V
M2: gm/id = 12.9612 S/A, Vstar = 0.154306 V
M3: gm/id = 7.39491 S/A, Vstar = 0.270456 V
M4: gm/id = 7.44034 S/A, Vstar = 0.268805 V
M5: gm/id = 7.03101 S/A, Vstar = 0.284454 V
M6: gm/id = 6.96085 S/A, Vstar = 0.287321 V
    
```

Is the current and g_m in the input pair exactly equal? Why?

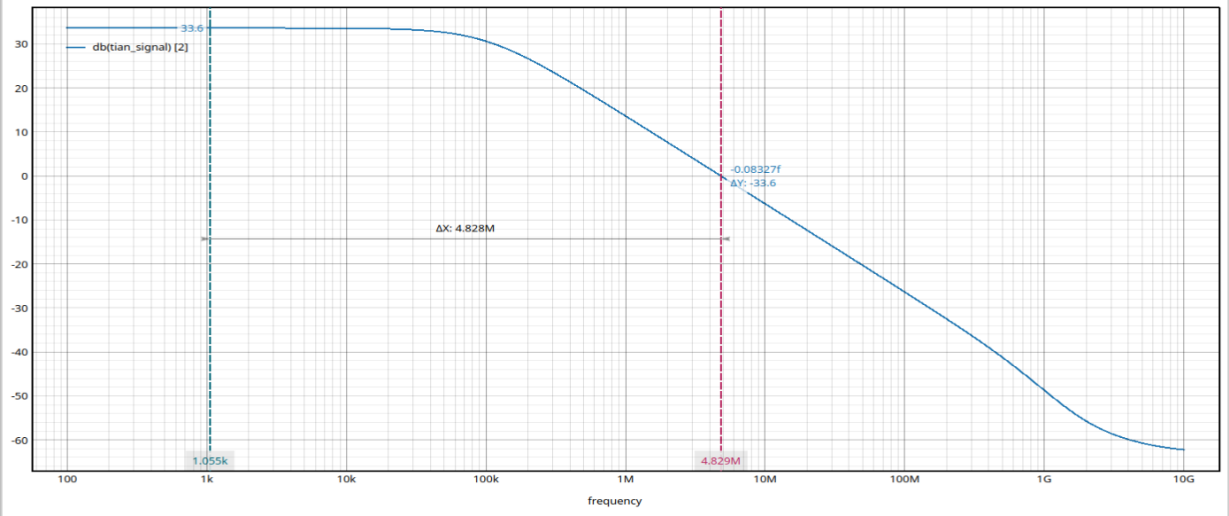
No, due to the systematic mismatch, because the feedback force the output to equal the input that make an offset voltage between input pair (mismatch between V_{GS1} and V_{GS2}) that cause mismatch between the 2 transistors.

Calculate the mismatch in I_D and g_m

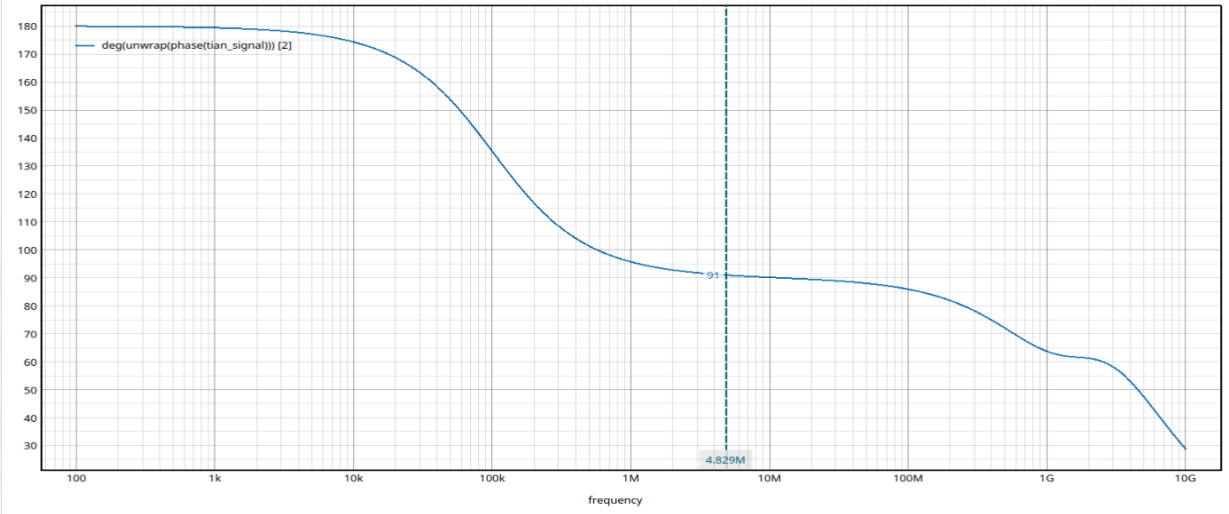
The mismatch in the current = $I_{D1} - I_{D2} = 13.04\mu A - 12.2\mu A = 0.84\mu A$

The mismatch in $g_m = g_{m1} - g_{m2} = 164\mu S - 158\mu S = 6\mu S$

Loop gain



Bode plot: LG mag(dB), GX=4.83MHz



Bode plot: LG phase, 89° phase margin

```
gain_crossover_freq = 4.828682e+06  
phaseatzerogain    = 9.099616e+01  
pm = 8.900384e+01
```

Compare LG DC gain and GX with those obtained from open-loop simulation

parameter	LG simulation	Open loop simulation
DC gain	33.6dB	34.25dB
GX (UGF)	4.83MHz	5MHz

Comment

From the results shown above we can conclude that, the system will be stable as it have 89° phase margin, and its transient response will be overdamped since phase margin > 76°”critical damped PM”

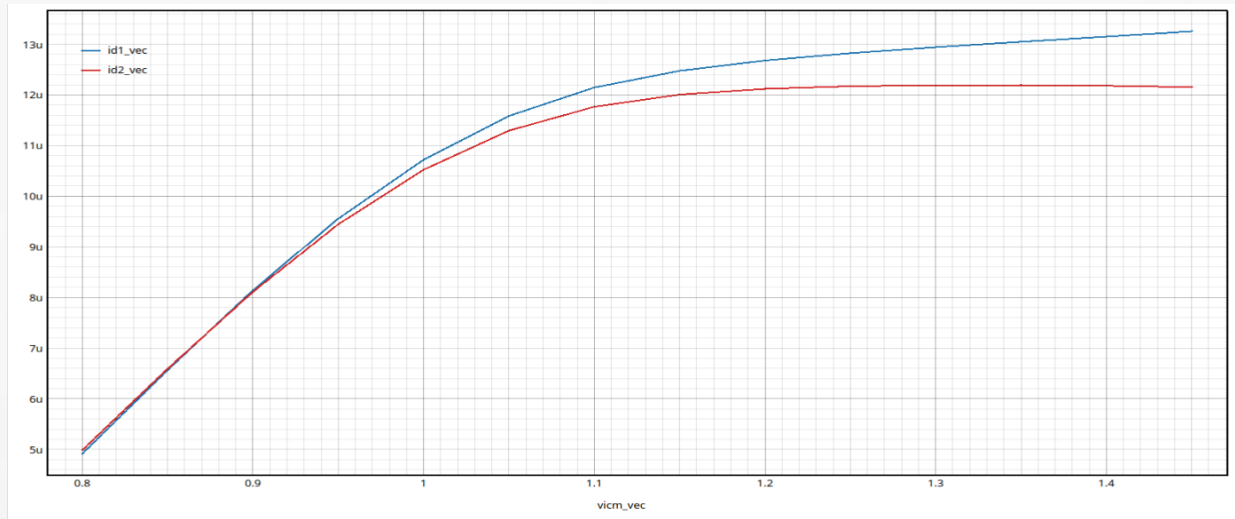
Compare simulation results with hand calculations in a table

parameter	simulation	Hand calculations
DC gain	33.6dB	34dB
GX (UGF)	4.83MHz	5MHz

Effect of Mismatch on CMRR

DC feedback loop will introduce mismatch in the input pair (why?)

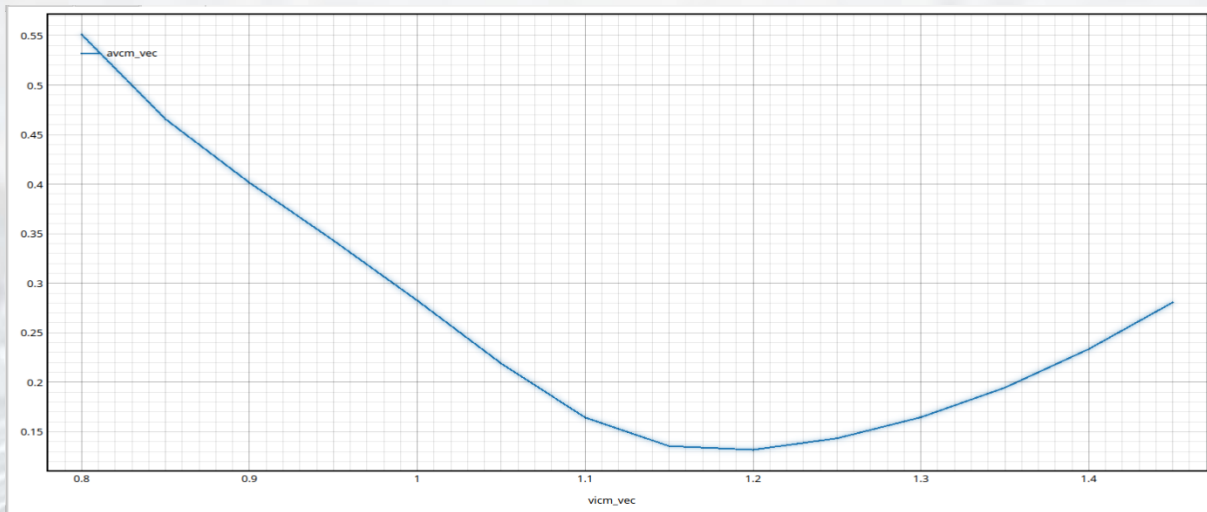
because the feedback force the output to equal the input that make an offset voltage between input pair (mismatch between V_{GS1} and V_{GS2}) that cause mismatch between the 2 transistors.



Comment:

Initially at small $V_{icm} \Rightarrow$ the feedback almost keeps two branches same

As V_{icm} increases $\Rightarrow$ the mismatch increases that make difference between the currents as $V_{icm} \uparrow \uparrow$



Comment:

Initially at small V_{icm} M_5 out of saturation $\Rightarrow$ small $R_{SS} \Rightarrow$ high A_{vcm}

As V_{icm} increases $\Rightarrow R_{SS}$ will increase $\Rightarrow A_{vcm}$ will decrease

Then at high V_{icm} , V_{out} has to follow input due to feedback that makes systematic mismatch Δg_m in input pair that cause degradation in CMRR $\Rightarrow A_{vcm} \uparrow \uparrow$

Bonus

Total area = $11.8\mu m^2$

Total current = $35.25\mu A$

GBW = $5MHz$

CMIR $\approx 0.641V$

FOM $\approx \frac{5MHz}{35.25\mu A \times 11.8\mu m^2} = 0.012 MHz/\mu A \cdot \mu m^2$

